

PitStop

Master System Documentation

Complete reference: customer app, owner panels, Super Admin, database, auth, booking, and deployment.

التوثيق الشامل — عميل، صاحب محل، مدير فرع، Super Admin، قاعدة البيانات، والحجز

[Expo 54](#)[React Native](#)[Supabase](#)[Super Admin](#)[AR / EN](#)[PitStop 2.0](#)[Premium SaaS](#)[Branch GPS](#)

Project: car-care-eg · Version 1.0.0 · Updated: Premium tiers, branch GPS sync, web multi-tab auth, El Rehab demo, wash service-only pricing

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1. Architecture Overview

نظرة عامة على البنية

Stack

Layer	Technology
Mobile / Web UI	Expo Router ~54, React Native, expo-linear-gradient, expo-image
State	React Context (auth, i18n, theme, shop catalog)
Local storage	AsyncStorage (bookings, extras, loyalty stamps, notifications, wash branches cache)
Legacy garage	SQLite via expo-sqlite (catalog, odometer — hidden tabs)
Backend	Supabase Auth + Postgres (optional; app works offline-first)
Push	Expo Notifications → shop_push_tokens (native only)
AI	OpenAI GPT-4o-mini (Assistant tab)

Provider tree (startup)

ThemePreferenceProvider → I18nProvider → ShopCatalogProvider
→ CustomerAuthProvider → AuthProvider → ShopAuthProvider → AppBootstrap

Data flow (hybrid)

Customer action (book, order)
→ lib/booking/storage.ts or partsStorage.ts
→ Supabase insert (if configured) + AsyncStorage cache
→ Owner notification (local + optional Expo push)
→ Owner approves/declines → customer in-app notification

Shop type enum

maintenance · wash · parts · accessories · winch

2. App Routes & Screens

المسارات والشاشات

Root stack (app/_layout.tsx)

Route	File	Purpose
/welcome	welcome.tsx	Customer + owner login, merchant registration, guest, language
/(tabs)/*	(tabs)/_layout.tsx	Main tab navigator
/service/[type]	service/[type]/index.tsx	Pick area (Cairo/Giza) for service type
/service/[type]/[areald]	service/[type]/[areald].tsx	Shops in area → profile or parts shop
/shop-profile/[shopId]	shop-profile/[shopId].tsx	Public profile: gallery, hours, services, reviews, book
/book/[shopId]	book/[shopId].tsx	Booking form: date, slots, car, phone, notes
/nearby/[type]	nearby/[type].tsx	GPS discovery — wash lists per branch, Haversine sort, refreshes on focus
/parts-shop/[shopId]	parts-shop/[shopId].tsx	Parts/accessories cart + checkout
/auth-required	auth-required.tsx	Gate before booking/purchase for guests
/settings/vehicles	settings/vehicles.tsx	Customer vehicle list
/shop/wash-owner-hub	shop/wash-owner-hub.tsx	Wash hub: notifications, orders, history
/admin	admin/index.tsx	Super Admin panel (dashboard, pending, merchants)
/reset-password	reset-password.tsx	Supabase password reset deep link
/login, /verify	login.tsx, verify.tsx	Legacy phone OTP (garage prototype)
/car/[id], /add-car	car/, add-car.tsx	Legacy Egypt garage (SQLite odometer)

Customer tabs (app/(tabs)/)

Tab	Visible	Purpose
index (Home)	Customer/guest	Service cards, offers, next booking
bookings	Customer/guest	My bookings, cancel, history
favorites	Logged-in customer	Saved shops
assistant	Customer/guest	Driver Network + AI chat
settings	Customer/guest	Language, theme, vehicles, support, logout
shop	Shop session	Owner login or owner dashboard
catalog, alerts	Hidden	Legacy garage features

Navigation rules (AppBootstrap.tsx)

- Public paths: /welcome, /reset-password, /auth-required, /login, /verify
- **Admin session** → redirect to /admin
- **Pending owner** → stay on /welcome?pending=1 (no shop access)
- Approved shop session → redirect to /shop

- No session on protected route → /welcome
- Customer on /welcome after login → home or returnUrl deep link

3. Authentication & Roles

المصادقة والأدوار

Roles (`public.user_role`)

Role	Login	shop_id	branch_id	Purpose
customer	Yes	—	—	Books services, favorites, notifications
owner	Yes	Required	—	Owns brand, all branches, creates managers
branch_manager	Yes	Required	Required	Runs one branch day-to-day
pending_owner	Yes	Required	—	New merchant awaiting admin approval; <code>is_active=false</code>
admin	Yes	—	—	Super Admin → <code>/admin panel</code>

Employees are NOT auth users — stored in `branch_employees` only.

Shop visibility: `shops.is_active=false` hides shop from customer catalog until admin approves.

Customer auth (`CustomerAuthContext.tsx`)

- Supabase email/password: `signIn`, `signUp`, `resetPassword`
- Guest mode: local flag, no Supabase session
- Blocks shop staff emails from customer login
- Requires email confirmed for login
- Storage: `@pitstop/customer-session`, `@pitstop/guest-session`

Shop auth (`ShopAuthContext.tsx`)

- Resolves from `public.users` (`owner` | `branch_manager` | `admin` | `pending_owner`) or legacy `shops.owner_email`
- Flags: `isOwner`, `isBranchManager`, `isAdmin`, `isPendingOwner`, `isPremium`
- `registerOwner()` — `signUp` + RPC `register_shop_owner`
- Login results: `ok` | `ok_admin` | `pending_approval` | `invalid_credentials` | `email_not_confirmed` | `email_login_disabled` | ...
- Storage: `@pitstop/shop-session` (staff email)
- Branch manager invite uses `staffInviteLock` to prevent session hijack
- **Web login fix:** `lib/auth/authMutationLock.ts` blocks `onAuthStateChange` side effects during `signIn/signUp` to prevent Supabase auth deadlocks (infinite "Signing in...")

Customer auth — web deadlock fix

Both `CustomerAuthContext` and `ShopAuthContext` defer Supabase queries out of `onAuthStateChange` using `setTimeout(0)` and skip listeners while `beginAuthMutation()` is active. Welcome screen shows contextual button labels: Signing in... / Creating account... / Redirecting...

إصلاح تعليق تسجيل الدخول على الويب: لا تُنفَّذ استعلامات Supabase داخل `onAuthStateChange` أثناء `signIn`.

Auth storage on web (per-tab sessions)

Web uses **tab-isolated** storage so you can test owner + customer in parallel browser tabs:

- `lib/supabase/authStorage.ts` — Supabase JWT in `sessionStorage` (not `localStorage`)
- `lib/storage/webTabAuthStorage.ts` — `@pitstop/customer-session`, `@pitstop/guest-session`, `@pitstop/shop-session` in `sessionStorage`

- `lib/supabase/webTabAuthIsolation.ts` — disables Supabase BroadcastChannel cross-tab sync; `signOut({ scope: 'local' })` on web so logout in one tab does not revoke the other tab's session
- Legacy Supabase keys in `localStorage` are cleared on first client init

على الويب: كل تاب له session مستقل — owner في تاب و customer في تاب ثاني من غير ما يتزامنوا.

Role hierarchy (updated)

Super Admin (platform)

- └ Approves / rejects `pending_owner` registrations
- └ Views platform stats (bookings, 12% fee)
- └ Sees all merchants + branches

Owner (approved)

- └ Creates branches
- └ Creates / links Branch Managers
- └ Sees all branches + reports

`pending_owner` (awaiting approval)

- └ Shop hidden from customers until admin accepts

Branch Manager (one branch)

- └ Manages daily operations
- └ Adds employees (no login)

Employees → roster only (`branch_employees`)

4. Customer Features

مميزات العميل

Home (/tabs)/index.tsx)

- Service entry cards: maintenance, wash, parts, accessories
- Live promotional offers from shops
- Next upcoming booking snippet
- Car profile shortcut
- **Loyalty card** — 5-stamp wash loyalty program (logged-in customers only)

Car Wash customer UX (new)

- **Busy badge:** pulsing red notice when owner sets branch status to Busy — on shop list cards and shop profile
- **Loyalty stamps:** offline-first; +1 stamp when a wash booking reaches done; reward coupon at 5 stamps
- **Today's booking progress:** step timeline on My Bookings for same-day appointments (Pending → Confirmed → In Progress → Done)

مميزات غسيل جديدة: شارة مزدحم، بطاقة ولاء 5 أختام، شريط تقدم للحجوزات اليوم.

Discovery

- **By area:** /service/{type} → pick Maadi, Nasr City, El Rehab, etc.
- **Nearby GPS:** /nearby/{type} — device location permission → Haversine distance sort (closest first). For wash, lists each active shop_branches row (not just shop brand). Screen reloads on focus so owner GPS updates appear without full app restart.
- **Directions:** shop profile + list cards → Google Maps deep link (<https://maps.google.com/?q=lat,lng>) using branch coordinates when available
- **Favorites:** saved shops tab (login required)
- **Wash pricing display:** customers see per-service prices on shop profile — no separate “base price” chip for wash shops

Shop profile

- Gallery, bilingual name/address, phone, directions, share
- Services list with prices and duration
- Reviews (ratings, comments, likes, owner replies)
- Offers / coupons display
- **Wash busy badge** when washShopStatus=busy (synced from owner branch status)
- More info section (owner-configured)
- Book appointment CTA → requires login (not guest)

My bookings

- List all bookings for customer phone/account
- **Progress timeline** for today's active wash/service bookings (replaces static badge)
- Cancel pending/confirmed bookings
- Clear history
- In-app notifications for approval, decline, reminders

Driver Network + AI (/tabs)/assistant.tsx)

- Community posts (local AsyncStorage)
- Embedded AI assistant (OpenAI — requires API key)

Settings

- Arabic / English toggle
- Light / dark theme
- Vehicle management link
- Support, privacy, sign out

5. Shop Types & Owner Panels

أنواع المحلات ولوحات التحكم

Panel routing (app/(tabs)/shop.tsx)

wash type → WashOwnerPanel.tsx (full CMS)
parts/accessories → Store owner UI (inventory + orders)
maintenance/winch → Generic owner UI (profile + bookings + PDF)

Generic owner panel (maintenance, winch)

Section	Features
Profile	Bilingual name, address, phone, email, more info; winch toggle (maintenance)
Gallery	Add/remove photos
Pricing	Default service price (EGP)
Schedule	Open/close, slot duration; Friday closed default
Offers	Bilingual promotional offers
Bookings	Active requests (confirm/cancel/done), history, date-range reports
PDF report	View report modal + save/share PDF
Notifications	Booking + parts order alerts

Store owner panel (parts, accessories)

Section	Features
Profile + gallery	Same as generic (no schedule/bookings)
Inventory	Add items: name, price, stock, image; ±1 stock
Orders	Confirm / cancel / mark shipped; 12% platform fee on orders

Wash owner panel (WashOwnerPanel.tsx) — PitStop 2.0 + Premium

Section	Owner	Branch Manager	Premium required?
Dashboard stats (today + pending)	All branches	Own branch	No (Free)
Dashboard stats (revenue, rating, customers)	All branches	Own branch	Yes
Branch status (open/closed/busy/vacation)	Per branch	Own branch	No
Analytics widgets	Per branch	Own branch	Yes
Branch switcher + add branch	Yes	No (fixed branch)	Yes (Free = single branch only)
Profile + GPS pin + gallery	Per branch	Own branch	No
Services CRUD (each service has price)	Per branch	Own branch	Yes
Weekly hours + vacation	Per branch	Own branch	Yes
Coupons	Per branch	Own branch	Yes
Bookings workflow	All branches	Own branch	No
Employees roster	All branches	Own branch	Yes

Create branch manager	Yes	No	Yes
Reviews moderation	Yes	Yes (own branch)	Yes
PDF reports	All	Own branch	Yes

Premium gate: PremiumFeatureGate.tsx dims locked sections; tap opens PremiumUpgradeModal (EN/AR). Flag from shops.is_premium via ShopAuthContext.isPremium.

الخطة المجانية: إحصائيات أساسية + حالة الفرع + الملف + المعرض + الحجوزات. Premium: فروع متعددة، خدمات، تحليلات، كويونات، مديرين، تقارير.

Service categories (wash)

exterior_wash · interior_cleaning · full_package · detailing · engine_cleaning · custom

Wash owner hub (/shop/wash-owner-hub)

Tab	Purpose
notifications	Wash notification inbox
orders	Approve/reject/update booking statuses
history	Past bookings; clear history

6. Booking Flow

مسار الحجز من البداية للنهاية

End-to-end workflow

```
Home / Nearby / Favorites
→ Shop profile (/shop-profile/[shopId])
→ [Guest?] → /auth-required → /welcome login
→ Book (/book/[shopId]?serviceId=...)
→ Select date + time slot + car + phone + notes
→ createBooking() → status: pending
→ Owner notification (+ optional push)
→ Owner: confirm → confirmed | decline → cancelled
→ Customer notification + reminders (60 min, 30 min before)
→ Owner: in_progress → done (or no_show)
```

Booking statuses

pending → confirmed → in_progress → done | cancelled | no_show

Pricing (lib/booking/storage.ts)

Shop type	Booking price source
wash	Selected service priceEgp from branch menu; fallback = first active service (no owner "base price" UI)
maintenance	Owner default service price in shop extras, or type default 650 EGP
winch	Type default 500 EGP
other	Type default 420 EGP

Platform fee: **12%** of service price (shown in owner PDF reports).

Time slots

Built by `buildSlotsForShopDate()` from owner weekly hours + existing bookings. Submit disabled until a slot is selected.

Persistence

- Primary: Supabase bookings table (when configured)
- Cache: AsyncStorage @pitstop/bookings/v1
- On Supabase insert failure → error shown to customer (RLS fix: step4 SQL)

7. Parts & Store Orders

طلبات قطع الغيار والإكسسوارات

```
/parts-shop/[shopId]
  → Browse inventory (store table)
  → Cart + shipping address + phone
  → createPartsOrder()
  → Owner notification + push
  → Owner: confirm / cancel / ship
  → Customer notification
```

Order statuses: pending → confirmed → shipped | cancelled

8. Database Schema

هيكل قاعدة البيانات

Entity diagram

```
auth.users (Supabase Auth)
└─ public.users (role, shop_id, branch_id)

public.shops (brand)
├─ shop_branches (locations, hours, gallery, status)
│   └─ branch_services
│       └─ branch_employees (no login)
├─ bookings (+ branch_id, service_id)
├─ store (parts/accessories inventory)
└─ parts_orders + parts_order_items

shop_push_tokens · garage_snapshots (legacy) · areas
```

Core tables

Table	Purpose
areas	Cairo/Giza districts + el-rehab (El Rehab City demo)
shops	Brand catalog; latitude/longitude; is_premium; legacy owner_email
users	App identity: customer, owner, branch_manager, admin
shop_branches	Physical branches: profile, weekly_hours, vacation_mode, branch GPS latitude/longitude
branch_employees	Staff roster (full_name, phone, job_title, notes)
branch_services	Per-branch service menu
bookings	Appointments with status, pricing, branch_id
store	Parts/accessories SKUs with stock_qty
parts_orders	Customer store orders
shop_push_tokens	Expo push tokens per shop

RLS helpers

is_shop_owner · is_branch_manager · can_manage_shop · can_manage_branch

9. Supabase Setup & Migrations

إعداد Supabase وملفات ال SQL

Fresh project

1. supabase/schema.sql
2. supabase/seed.sql — 12 demo shops
3. supabase/seed-branches.sql — optional 2nd branch + employees

Existing project (PitStop 2.0 migration)

1. apply-pitstop-2.0-step1-enums.sql — run alone first
2. apply-pitstop-2.0-step2-main.sql — users, branches, employees
3. apply-pitstop-2.0-step3-branch-manager-rpc.sql — assign_branch_manager RPC
4. apply-pitstop-2.0-step4-bookings-rls.sql — restore booking insert policies
5. apply-pitstop-2.0-step5-admin-approval.sql — Super Admin, pending_owner, approve/reject RPCs
6. apply-pitstop-2.0-step6-shop-premium.sql — shops.is_premium boolean default false
7. seed-admin-user.sql — link Auth user as platform admin in public.users
8. seed-el-rehab-wash.sql — El Rehab area + shop-wash-rehab branch at 30.0244, 31.4939 + owner link (requires Auth user rehab.wash@demo.com first)
9. SETUP-AUTH-CHECKLIST.md — step-by-step auth setup (Email provider, demo accounts)
10. seed-test-data.sql — optional demo manager link

Other migration files

- migrate-roles-branches.sql — legacy → 2.0
- migrate-to-store.sql — spare_parts → store
- apply-pitstop-2.0-on-existing-db.sql — combined path

Step 5 RPCs (Super Admin)

RPC	Caller	Effect
register_shop_owner(...)	New merchant (authenticated)	Creates inactive shop + main branch; user → pending_owner
approve_shop_owner(user_id, shop_id)	Admin	user → owner, is_active=true; shop → is_active=true (live)
reject_shop_owner(user_id, shop_id)	Admin	Deletes shop; user → customer

Link owner after Auth user created

```
update public.users
set role = 'owner', shop_id = 'shop-wash-nile'
where email = 'wash@demo.com';
```

10. Key Library Modules

أهم ملفات /lib

Module	Role
lib/booking/storage.ts	Bookings CRUD, Supabase + AsyncStorage merge
lib/booking/catalogRepository.ts	Areas/shops from Supabase + cache
lib/booking/shopExtrasStorage.ts	Owner profile, schedule, services, offers, washShopStatus sync
lib/booking/loyaltyStampsStorage.ts	Offline loyalty stamps + reward coupons (@pitstop/loyalty_stamps)
lib/booking/shopSchedule.ts	Weekly hours, slot builder, open-now, busy status labels
lib/booking/reporting.ts	Date ranges, EGP, buildOwnerReportHtml()
lib/booking/commerceEvents.ts	Owner/customer notifications
lib/booking/bookingReminders.ts	60/30 min reminders (native)
lib/booking/partsStorage.ts	Store inventory + orders
lib/booking/wash/washBranchStorage.ts	Multi-branch state (Supabase-first + cache)
lib/booking/nearby.ts	Haversine distance, formatDistanceAway, shop sort helpers
lib/booking/washBranchNearby.ts	Lists wash branches from Supabase, sorts closest-first per branch pin
lib/linking/contact.ts	Phone, maps, openBranchDirections(), openMapsAtCoordinates()
lib/booking/wash/branchRepository.ts	Branch CRUD, GPS update, resolveRemoteBranchId, fetchDefaultBranchCoordinates
lib/booking/wash/branchManagerRepository.ts	Manager account RPC + signUp flow
lib/shop/ownerRegistrationRepository.ts	Merchant signUp + register_shop_owner RPC
lib/admin/adminRepository.ts	Pending list, merchants, stats, approve/reject
components/admin/AdminPanel.tsx	Super Admin UI (3 tabs)
lib/shop/shopStaffUser.ts	Resolve owner/manager/admin/pending from users
lib/supabase/client.ts	Supabase singleton; calls prepareWebTabAuthIsolation() on init
lib/supabase/authStorage.ts	sessionStorage (web) / AsyncStorage (native) for Supabase JWT
lib/supabase/webTabAuthIsolation.ts	Disable BroadcastChannel; local signOut; clear legacy localStorage keys
lib/storage/webTabAuthStorage.ts	Per-tab customer/guest/shop session keys on web
components/owner/PremiumFeatureGate.tsx	Locks premium wash owner sections
components/owner/PremiumUpgradeModal.tsx	Upgrade CTA modal (EN/AR)
lib/push/shopPush.ts	Expo push for owners (native)
lib/i18n/strings.ts	~928 translation keys × EN + AR
lib/auth/authMutationLock.ts	Prevents Supabase auth deadlock during signIn/signUp
components/ui/WashBusyBadge.tsx	Pulsing red busy notice for customers

components/customer/LoyaltyCard.tsx	5-stamp loyalty card on Home
components/customer/BookingProgressTimeline.tsx	4-step booking progress for today
lib/ui/userAlert.ts	window.alert on web (Alert.alert fallback)
lib/ai-chat.ts	OpenAI assistant

11. Internationalization

العربية والإنجليزية

- Locales: en | ar — stored in @pitstop/locale
- Default from device language (expo-localization)
- API: `t(key)`, `tp(key, vars)`, `isRTL`
- RTL: `I18nManager` (native) + `document.dir` (web)
- Toggle: Welcome screen + Settings

12. Notifications & PDF Reports

الإشعارات وتقارير PDF

Owner push (native only)

- Table: shop_push_tokens
- Triggers: new booking, new parts order
- Via Expo Push API

Customer in-app notifications

Kinds: booking_approved, booking_declined, booking_reminder, parts_order_confirmed, parts_order_cancelled

PDF reports

- `buildOwnerReportHtml()` — shop header, date range, gross/fee/net
- Web: scrollable modal preview → Save as PDF (browser print)
- Native: `printToFileAsync` + share
- Used in generic shop panel + `WashOwnerPanel`

13. Demo Accounts

حسابات تجريبية — كلمة السر: demo123

Platform admin

admin@demo.com / demo123

1. Supabase → Authentication → Providers → Email → **Enable Email provider**
2. Add user with **Auto Confirm User** checked
3. Run seed-admin-user.sql (or step5 + manual UPDATE)
4. Login: /welcome → Owner tab → redirects to /admin

See supabase/SETUP-AUTH-CHECKLIST.md for full troubleshooting.

```
update public.users set role = 'admin', is_active = true, shop_id = null
where email = 'admin@demo.com';
```

Shop owners (pre-approved in seed)

Email	Shop ID	Type
wash@demo.com	shop-wash-nile	wash (set is_premium=true in SQL to test Premium UI)
rehab.wash@demo.com	shop-wash-rehab	wash — El Rehab GPS demo (create Auth user + run seed-el-rehab-wash.sql)
maintenance@demo.com	shop-maint-autofix	maintenance
parts@demo.com	shop-parts-nasr	parts
accessories@demo.com	shop-accessories-maadi	accessories
winch@demo.com	shop-winch-maadi	winch

Also: wash2@, wash3@, maintenance2@, maintenance3@, parts2@, accessories2@, winch2@ — see seed.sql

Branch manager

manager.wash@demo.com / demo123 — Nile Auto Wash main branch

Customers

customer@demo.com · sara@demo.com / demo123

14. Environment & Running the App

تشغيل المشروع

Environment variables (.env)

```
EXPO_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
EXPO_PUBLIC_OPENAI_API_KEY=sk-... # for AI assistant
```

Fallback (app.json extra)

Supabase URL and anon key also in app.json → extra if .env unset.

Commands

```
cd car-care-eg
npm install
npm start          # Expo (offline mode)
npm run web        # http://localhost:8082
npm run android
npm run ios
```

Supabase project (demo)

URL: <https://qlopvpyeawauepsyrirz.supabase.co>

15. Limitations & Roadmap

القيود والتطوير المستقبلي

Current limitations

Limitation	Detail
Hybrid storage	Many features use AsyncStorage; Supabase when configured
Guest cannot book	Login required despite guest browsing
Merchant approval	New owners need admin approve before shop goes live
Loyalty coupons	Generated locally; not yet validated at checkout in book flow
Winch not on home	In DB but no home service card
Web push disabled	Owner push + reminders native only
Payments	No online payment integration yet
Legacy garage	Parallel SQLite app (hidden tabs)

Future enhancements (from docs)

- Online payments (Paymob/Fawry)
- Apply loyalty coupon codes at booking checkout
- Subscriptions
- Driver network on Supabase + moderation
- Vehicle cloud sync
- Multi-region (GCC expansion)
- Assign employees to bookings/shifts
- QR check-in, live queue ETA, fleet management

Implemented: loyalty stamps, today booking progress, wash busy badge, Super Admin, Premium SaaS tiers, branch GPS sync, web multi-tab auth, wash service-only pricing.

Related documents in repo

- docs/SCHEMA.md — database reference
- docs/FUTURE_ENHANCEMENTS.md — product roadmap
- docs/pitstop-2.0-owner-srs.html — Owner/Branch SRS
- pitstop2.0.pdf — Customer-side SRS (original)

16. UI Architecture & Design System

بنية الواجهة ونظام التصميم

Layout model

```
app/_layout.tsx (Root Stack)
├─ welcome, auth-required, book, shop-profile, nearby, parts-shop ...
└─ (tabs)/_layout.tsx (Bottom tabs – customer OR shop mode)
    ├─ index (Home)
    ├─ bookings | favorites | assistant | settings ← customer area
    └─ shop ← owner dashboard (tabs hidden when shop logged in)
```

Two UI modes

Mode	Trigger	Tabs shown	Default route
Customer / Guest	No shop session	Home, Bookings, Favorites*, Assistant, Settings	/ (Home)
Shop staff	ShopAuthContext.shop set	Shop tab only (bar hidden)	/shop → AppBootstrap forces /shop

*Favorites tab hidden for guests — login required.

Theme (constants/Theme.ts + ThemePreferenceContext)

- **Light / Dark** — user choice in Settings; persisted
- **Tokens:** bg, bgElevated, card, border, text, textMuted, accent, warm, onAccent
- **RTL:** Arabic flips layout via I18nContext + I18nManager
- **Owner wash accent:** warm color in dark mode for owner buttons

UI patterns used everywhere

Pattern	Where	Behavior
Pressable + gradient btn	welcome.tsx	Login/register; pointerEvents none on gradient (web fix)
OwnerSectionCard	WashOwnerPanel	Collapsible section with title + subtitle
Modal confirm	shop.tsx, WashOwnerPanel	Replaces Alert.alert on web for booking decisions
userAlert	welcome, book, shop	window.alert on web when Alert invisible
ScrollView sections	All owner panels	Long vertical CMS layout
iframe report preview	shop.tsx, WashOwnerPanel	Scrollable HTML report before PDF

17. Master Navigation Flowcharts

مخططات التنقل الرئيسية

17.1 App entry & auth routing

```
[App launch]
|
v
[AppBootstrap waits: CustomerAuth + ShopAuth ready]
|
+-- isAdmin? -----> router.replace('/admin')
|
+-- isPendingOwner? -----> /welcome?pending=1 (blocked from /shop)
|
+-- approved shop session? --> router.replace('/shop')
|
+-- no session + protected? --> /welcome
|
+-- customer/guest on /welcome? --> / or returnTo
|
v
[/ (tabs) or stack screens]
```

17.2 Customer booking journey

```
Home (/ (tabs)/index)
| tap service card (maintenance|wash|parts|accessories)
v
/service/[type] → pick area (Maadi, Nasr City...)
v
/service/[type]/[areaId] → shop list
|
+-- parts/accessories --> /parts-shop/[shopId]
|
+-- maintenance/wash --> /shop-profile/[shopId]
|
| [Book button]
v
guest? --> /auth-required --> /welcome?returnTo=/book/...
|
v
/book/[shopId]?serviceId=...
| fill phone, car, date, time slot, notes
v
createBooking() --> success modal --> /bookings tab
```

17.3 Alternative discovery paths

```
Home --search/filter--> service cards
Home --live offers--> shop-profile
Nearby GPS (/nearby/[type]) --> shop-profile | parts-shop
Favorites tab --> shop-profile (login required)
Shop profile: Call | Directions | Share | Favorite | Gallery | Reviews
```

17.4 Owner login & panel routing

```
/welcome → Owner tab
|
```



```

+-- Login (approved owner/manager)
|   v
|   ShopAuthContext.login() → ok
|   v
|   router.replace('/shop') → WashOwnerPanel | store UI | generic UI
|
+-- Login (admin)
|   v
|   router.replace('/admin') → AdminPanel
|
+-- Login (pending_owner)
|   v
|   pending_approval message – no shop access
|
+-- Register as merchant
|   v
|   registerOwner() → signUp + register_shop_owner RPC
|   v
|   pending_owner + shop is_active=false → wait for admin

```

17.5 Wash owner hub (secondary navigation)

```

WashOwnerPanel → Hub shortcuts
  v
/shop/wash-owner-hub?tab=notifications|orders|history
|
+-- notifications: wash notification inbox
+-- orders: approve/reject/in_progress/done bookings
+-- history: completed/cancelled list + clear

```

17.6 Branch manager vs owner (wash UI)

```

Same login form (/welcome Owner tab)
  v
ShopAuthContext resolves role
|
+-- isOwner --> branch tabs visible, create branch, create manager
|
+-- isBranchManager --> single branch, no manager section, bookings filtered by branch_id

```

18. Customer UI — Module by Module

واجهة العميل — شرح كل جزء

Module A — Welcome / Auth (app/welcome.tsx)

UI elements: Logo, language toggle, Customer | Owner mode tabs, guest button, form box.

Customer tab: Login ↔ Register toggle, email/password, optional vehicles on register, forgot password link.

Owner tab: Shop email + password, demo accounts hint, merchant registration link.

Actions: onCustomerSubmit → login/register; onOwnerSubmit → shop login (owner or admin); continueAsGuest → home without Supabase.

Feedback: formMessage inline (red) + userAlert on errors; button shows Signing in... / Creating account... / Redirecting... (not stuck on "Working...").

Admin routing: ok_admin → /admin; pending owner → blocked message on welcome.

شاشة الدخول: تسجيل عميل/صاحب محل، ضيف، وإعادة توجيه الأدمن — مع إصلاح تعليق تسجيل الدخول على الويب.

Module B — Home (app/(tabs)/index.tsx)

Sections top→bottom:

1. Header logo + sign out (if logged in)
2. **Loyalty card** (LoyaltyCard) — 5 car-icon stamps; visible for logged-in customers only
3. Car profile card — saved make/model; edit modal
4. Next booking card — upcoming appointment or empty state
5. Live offers carousel — wash/maintenance promotions
6. Service search + filter (all / wash only)
7. 4 service cards → navigate to /service/{type}

Loyalty reward modal: when 5th stamp unlocks, Home shows modal with coupon code LOYAL-XXXXXX (100% off, 90 days). Stamps sync on focus via syncLoyaltyFromBookings().

الرئيسية: بطاقة ولاء 5 أختام + نافذة كوبون عند اكتمال البرنامج.

Module C — Area & Shop List (app/service/)

Step 1: Grid of Cairo/Giza areas with shop count.

Step 2: Shops in selected area; card shows name, rating, distance hint, open status.

Wash busy badge: on ShopListCard, when shop type is wash and extras washShopStatus === 'busy', shows pulsing red WashBusyBadge (compact pill).

Routing: parts/accessories → parts-shop; others → shop-profile.

قائمة المحلات: شارة «مزدحم» حمراء نابضة لمحلات الغسيل عندما يضبط المالك الحالة Busy.

Module D — Shop Profile (app/shop-profile/[shopId].tsx)

UI block	User action
Hero image + gallery swipe	View photos
Name, address, phone	Call, open maps

Schedule line	See open hours
Busy notice	WashBusyBadge full-width when wash + busy status
Services list	Tap service → book with serviceId
Offers	Read promotions
More info	Owner free-text details
Reviews	Read/submit rating (logged in)
Share / Favorite	Social share; save shop
Book CTA (sticky)	→ /book or auth-required

Module E — Booking Form (app/book/[shopId].tsx)

Field	Source	Validation
Phone	Customer profile	Egypt E.164
Car type / color	Vehicle storage or manual	Car type required
Date picker	BookingDatePicker	Owner schedule days
Time slots grid	buildSlotsForShopDate	Required; booked slots disabled
Service summary	Selected service or shop default	Price + duration shown
Notes	Free text	Optional
Submit	createBooking	Disabled if no slot

Success: Modal → View my bookings → /bookings tab.

Module F — My Bookings (app/(tabs)/bookings.tsx)

List cards: shop name, service, datetime, status badge, cancel button (pending/confirmed).

Today progress timeline: for bookings scheduled *today* with status in pending | confirmed | in_progress | done, the static status badge is replaced by BookingProgressTimeline (4 steps with accent connectors).

Clear history action for done/cancelled items.

حجوزاتي: شريط تقدم مباشر للحجوزات اليوم بدل الشارة الثابتة.

Module G — Favorites, Assistant, Settings

Favorites: Saved shop cards → shop-profile. Login required.

Assistant: Driver Network posts (local) + AI chat panel (OpenAI key required).

Settings: Language EN/AR, theme light/dark, vehicles link, support links, privacy, sign out.

Vehicles: /settings/vehicles — add/edit/delete customer cars.

Module H — Parts Shop (app/parts-shop/[shopId].tsx)

Product grid, cart badge, qty +/-, shipping address, phone, checkout → createPartsOrder.

No time slots — inventory-based e-commerce flow.

Module I — Nearby (`app/nearby/[type].tsx`)

Requests GPS permission → sorts shops by distance.

Filter chips: All, Top rated, Price, Distance, Open now, Favorites.

Module J — Customer Notifications (`CustomerNotificationsBell`)

Bell icon in tab headers (logged-in customer only).

Shows booking approved/declined/reminder + parts order updates.

Processes due reminders on focus.

19. Owner UI — Module by Module

واجهة صاحب المحل — شرح كل جزء

19.1 Generic owner panel (app/(tabs)/shop.tsx — maintenance / winch)

Header bar

Profile cover image, notifications bell, logout. Shop name from catalog.

Profile section

Fields: profile name EN/AR, address EN/AR, phone, email, more info EN/AR.

Maintenance only: Winch service toggle (shows winch add-on to customers).

Save profile info button → shopExtrasStorage.

Photo album

Add image URL, remove image. Gallery shown on shop-profile.

Service price + schedule

Default price EGP, open/close time, slot duration (15+ min). Friday closed default.

Generates customer time slots on book screen.

Offers

Add bilingual offer title/body, expiry. Cancel active offers.

Bookings dashboard

Report: date from/to, quick last N days, preview list, View report modal, Save PDF.

Active requests: pending + confirmed — Confirm | Cancel | Mark done buttons via confirm modal.

History: done + cancelled — show/hide toggle.

19.2 Store owner panel (parts / accessories)

Same profile + gallery as generic. **No schedule/bookings.**

Inventory table: add item (name, price, stock, image URL), +/- stock buttons.

Orders table: customer phone, items, total, status — Confirm | Cancel | Shipped.

19.3 Wash owner panel (WashOwnerPanel.tsx) — section order

#	Section (i18n key)	UI controls	Owner only?
1	wash_dashboard_*	Stat cards: today bookings, pending, revenue, rating, customers, top service	Stats scope differs
2	wash_status_*	Open / Closed / Busy / Vacation + vacation dates	No
3	wash_analytics_*	Revenue & booking charts (computed locally)	No

4	wash_hub_*	3 shortcut buttons → wash-owner-hub tabs	No
5	wash_branch_*	Branch tabs, add branch, rename	Owner only
6	wash_profile_*	Branch profile fields + GPS capture button (syncs to Supabase immediately) + save	No
7	wash_gallery_*	Image URL list add/remove	No
8	wash_services_*	Service list, add/edit modal (EN/AR, price, duration, category, visible, sort)	Premium
9	wash_hours_*	Day picker 0–6, open/close/break, copy to all days	Premium
10	wash_coupons_*	Coupon list, add percent/fixed code	Premium
11	wash_manager_*	Link email Create login (name, email, password)	Owner + Premium
12	wash_employees_*	Employee list per branch, add/edit/delete roster	Premium
13	wash_reviews_*	Hide review, reply, report	Premium
14	wash_report_*	Date range report + PDF (same engine as generic)	Premium
15	Active bookings inline	Status workflow: accept, reject+note, in progress, done, no show	Branch filtered · Free

19.4 Wash owner hub screens

wash-owner-hub.tsx

Tab: notifications → list from washNotificationCenter (unread badge)

Tab: orders → booking cards + action buttons + call customer

Tab: history → past bookings + delete/clear

19.5 Super Admin panel (AdminPanel.tsx)

Tab	Module	Actions
Dashboard	Platform stats cards	Read-only metrics from bookings + users
Pending	PendingOwnerRequest list	Accept / Reject → Supabase RPC
Merchants	ActiveMerchant list	View approved owners + branch count

Entry: /welcome Owner tab → admin login → /admin. Uses OwnerSectionCard for sections.

20. Screen-by-Screen Reference

مرجع سریع لكل شاشة

/welcome Auth gateway. Customer register/login, guest, owner login. Language toggle.	/(tabs)/index Home hub. Loyalty card, services, offers, next booking, car profile.
/service/[type] Pick geographic area.	/service/[type]/[areald] Shop list for area.
/shop-profile/[shopId] Public shop page: services + prices, directions, busy badge, book.	/book/[shopId] Appointment form + slot picker.
/bookings tab Customer appointments + today progress timeline.	/favorites tab Saved shops (login).
/assistant tab Driver network + AI chat.	/settings tab App preferences.
/settings/vehicles Customer car list CRUD.	/nearby/[type] GPS shop discovery.
/parts-shop/[shopId] Storefront + cart checkout.	/auth-required Login gate with returnTo.
/(tabs)/shop Owner dashboard (type-specific panel).	/shop/wash-owner-hub Wash notifications/orders/history.
/admin Super Admin: dashboard, pending merchants, active merchants.	/reset-password Supabase password reset handler.

Tab visibility matrix

Tab	Guest	Customer	Shop owner
Home	✓	✓	hidden
Bookings	✓	✓	hidden
Favorites	hidden	✓	hidden
Assistant	✓	✓	hidden
Settings	✓	✓	hidden
Shop	hidden*	hidden	✓ (only tab)

*Shop tab visible when no customer area and not logged in as customer — rare edge case.

21. Shared UI Components

المكونات المشتركة

Component	Path	Used for
AppBootstrap	components/AppBootstrap.tsx	Auth routing, splash hide
OwnerSectionCard	components/owner/OwnerSectionCard.tsx	Wash owner CMS sections
ServiceOptionCard	components/ui/ServiceOptionCard.tsx	Home service type cards
BookingDatePicker	components/ui/BookingDatePicker.tsx	Book screen date selection
CustomerNotificationsBell	components/customer/CustomerNotificationsBell.tsx	In-app notification dropdown
WashBusyBadge	components/ui/WashBusyBadge.tsx	Pulsing red busy notice (shop list + profile)
LoyaltyCard	components/customer/LoyaltyCard.tsx	5-stamp loyalty card on Home
BookingProgressTimeline	components/customer/BookingProgressTimeline.tsx	4-step booking progress (My Bookings today)
ShopCatalogProvider	context/ShopCatalogContext.tsx	Loads areas/shops cache for UI lists
userAlert	lib/ui/userAlert.ts	Cross-platform error dialogs

Context → UI binding

Context	UI reads	UI effect
I18nContext	t(), locale, isRTL	All labels; remount tabs on locale change
ThemePreferenceContext	useAppTheme()	Colors on every screen
CustomerAuthContext	customer, isGuest, busy	Auth gates, profile prefill
ShopAuthContext	shop, isOwner, isBranchManager, isPremium	Panel type, branch tabs, premium gates
ShopCatalogContext	ready, version	Shop lists, home offers

22. UI State & Context Flow

تدفق الحالة بين الواجهة والبيانات

22.1 Customer books — UI ↔ data

```
[Book screen state: phone, carType, dateYmd, timeSlot, notes]
  |
  | onSubmit
  v
createBooking() → Supabase bookings + AsyncStorage cache
  |
  +→ pushOwnerNotification (owner bell)
  +→ pushWashCenterNotification (wash hub)
  +→ sendShopPushForBooking (native push)
  |
  v
[Success modal] → user taps → /bookings tab reloads listBookingsForPhone()
```

22.2 Owner confirms — UI ↔ data

```
[shop.tsx or wash-owner-hub booking card]
  |
  | Confirm button → confirm modal
  v
updateBookingStatus(id, 'confirmed')
  |
  +→ pushCustomerNotification('booking_approved')
  +→ scheduleBookingReminders (60m, 30m)
  |
  v
[CustomerNotificationsBell shows approval]
[Bookings tab status badge updates on refresh]
```

22.3 Wash branch switch — UI ↔ data

```
[Owner taps branch tab in WashOwnerPanel]
  |
  v
setActiveBranch(branch) + useEffect reload
  |
  +→ loadBranchEmployees(branch.id) from Supabase
  +→ applyBranchToForms() fills profile/gallery/hours fields
  |
  v
[All section forms show selected branch data]
[Save writes back to washBranchStorage → branchRepository Supabase sync]
```

22.4 Report PDF — UI ↔ data

```
[Owner sets date range + Apply]
  |
  v
filterBookingsByDateRange() from listBookingsForShop()
  |
  v
buildOwnerReportHtml() → HTML string
  |
```

+— Web: setReportPreviewHtml → Modal iframe (scroll) → Save PDF = print
+— Native: Print.printToFileAsync → Sharing.shareAsync

22.5 Local vs cloud (what UI shows when offline)

UI area	Offline behavior
Shop lists	Cached catalog from last Supabase refresh
Bookings list	AsyncStorage merge; may differ from cloud
Owner extras (schedule, offers)	Local shopExtrasStorage until synced
Wash branches	Local cache + Supabase when online
Loyalty stamps	Fully offline AsyncStorage; stamps on wash booking done
Wash busy status	Owner sets in panel → shopExtrasStorage → customer badge
Login	Requires Supabase (shows not_configured if missing)
Admin panel	Requires Supabase + step5 RPCs; read-only offline

22.6 Loyalty stamp — UI ↔ data

```
[Owner marks wash booking → done]
  |
  v
updateBookingStatus() in lib/booking/storage.ts
  |
  v
recordWashBookingDone(booking, previousStatus)
  |
  +→ AsyncStorage @pitstop/loyalty_stamps (per customer id/phone)
  +→ +1 stamp if wash + first time for booking id
  +→ at 5 stamps: generate LOYAL-XXXXXX coupon, reset stamps to 0
  |
  v
[Customer opens Home tab]
  |
  v
syncLoyaltyFromBookings() backfills any missed done bookings
consumePendingLoyaltyReward() → reward modal with coupon code
```

22.7 Wash busy status — owner ↔ customer

```
[WashOwnerPanel → Branch status → Busy]
  |
  v
saveWashBranchStatus() → washBranchStorage + syncWashBranchToShopExtras()
  |
  v
shopExtrasStorage.washShopStatus = 'busy' (preserved in normalizeExtras)
  |
  v
ShopCatalogContext refresh → ShopListCard + shop-profile read extras
  |
  v
[Customer sees WashBusyBadge – pulsing red notice]
```

23. Super Admin — Overview & Roles

لوحة Super Admin — نظرة عامة

PitStop 2.0 adds a **merchant approval gate**: new shop owners cannot go live until a platform admin approves them. Existing seed owners (e.g. wash@demo.com) remain pre-approved.

State machine — merchant account

State	users.role	users.is_active	shops.is_active	Customer catalog	App route
New registration	pending_owner	false	false	Hidden	/welcome (pending message)
Approved	owner	true	true	Visible	/shop
Rejected	customer	true	deleted	—	/ (customer area)

Admin capabilities

- View platform-wide booking stats and 12% platform fee estimate
- List pending merchant registration requests
- Accept → atomic RPC activates user + shop
- Reject → deletes shop row, resets user to customer
- List active merchants with branch counts

24. Super Admin — Flowcharts

مخططات Super Admin والتسجيل

24.1 Merchant self-registration

```
/welcome → Owner tab → "Register as merchant"
|
v
Fill: business name, owner name, phone, address, shop type, area, email, password
|
v
ShopAuthContext.registerOwner()
|
+-- supabase.auth.signUp({ email, password, metadata })
|
+-- RPC register_shop_owner(...)
|   |
|   +-- INSERT shops (is_active=false, owner_user_id)
|   +-- INSERT shop_branches (slug=main)
|   +-- UPDATE users SET role=pending_owner, is_active=false, shop_id
|
v
Show: "Pending approval" (owner cannot access /shop yet)
|
v
[Admin reviews in /admin → Pending tab]
```

24.2 Admin approval flow

```
Admin logs in (admin@demo.com) via Owner tab on /welcome
|
v
AppBootstrap → router.replace('/admin')
|
v
AdminPanel → Pending requests tab
|
+-- [Accept] → RPC approve_shop_owner(user_id, shop_id)
|   |
|   +-- users: role=owner, is_active=true
|   +-- shops: is_active=true
|   |
|   v
|   Shop appears in customer catalog (refreshCatalog)
|   Owner can login → /shop
|
+-- [Reject] → RPC reject_shop_owner(user_id, shop_id)
|   |
|   +-- DELETE shop_branches + shops
|   +-- users: role=customer, shop_id=null
|   |
|   v
|   Registration removed; user stays as customer
```

24.3 Catalog visibility

```
refreshCatalog() queries shops WHERE is_active = true
|
```

```

  v
pending_owner shops (is_active=false) NEVER shown to customers
|
  v
After approve_shop_owner → shop included on next catalog refresh

```

24.4 Platform stats calculation

```

AdminPanel → Dashboard tab
|
  v
fetchPlatformStats() in adminRepository.ts
|
+-- COUNT all bookings
+-- FILTER status = 'done' for revenue
+-- SUM platform_fee_egg OR 12% of service_price_egg
+-- COUNT pending_owner users
+-- COUNT active owner users
|
  v
Display stat cards (OwnerSectionCard layout)

```

25. Super Admin — Modules & UI

شرح وحدات لوحة المشرف

Route: `app/admin/index.tsx` + `components/admin/AdminPanel.tsx`

Full-screen layout (web-optimized): fixed header + horizontal tab chips + scrollable content.

Tab A — Dashboard (`admin_tab_dashboard`)

Stat card	Data source
Total bookings	COUNT bookings
Completed	status = done
Gross revenue	SUM service_price_egp (done)
Platform fee (12%)	SUM platform_fee_egp or calculated
Pending owners	COUNT users role=pending_owner
Active merchants	COUNT users role=owner, is_active

Tab B — Pending requests (`admin_tab_pending`)

Each card shows: shop name, type, owner email, phone, address.

Accept → `approveShopOwner()` → RPC

Reject → `rejectShopOwner()` → RPC

Uses `userAlert` for success/error on web.

Tab C — Active merchants (`admin_tab_merchants`)

Lists approved owners: shop name, type, email, active branch count from `shop_branches`.

Header actions

- Title: Super Admin / لوحة المشرف
- Signed-in email from `ShopAuthContext.staff`
- Sign out → `useAppSignOut()` → `/welcome`

26. Merchant Registration Module

وحدة تسجيل التاجر الجديد

UI: `app/welcome.tsx` — **Owner tab**

Toggle: Login ↔ Register as merchant (`owner_register_link`)

Register fields:

1. Business name (`shopName`)
2. Owner full name (`name`)
3. Egypt mobile phone — validated
4. Business address
5. Shop type chips: wash, maintenance, parts, accessories
6. Area chips: from catalog areas (Maadi, Nasr City, ...)
7. Email + password (strong password rules)

Submit: `registerOwner()` in `ShopAuthContext`

Backend: `lib/shop/ownerRegistrationRepository.ts`

1. `supabase.auth.signUp` with metadata (`name`, `phone`, `account_type=shop_owner`)
2. `supabase.rpc('register_shop_owner', { p_shop_name, p_shop_type, p_area_id, ... })`
3. On RPC failure → `signOut` and return invalid
4. On success → session restored as `pending_owner`

Login while pending

Credentials valid but `login()` returns `pending_approval`.

AppBootstrap keeps user on `/welcome` with pending message — no access to `/shop` or customer booking as owner.

27. New Library Modules (Admin)

ملفات الكود الجديدة

File	Exports / purpose
lib/admin/adminRepository.ts	listPendingOwnerRequests, listActiveMerchants, fetchPlatformStats, approveShopOwner, rejectShopOwner
lib/shop/ownerRegistrationRepository.ts	registerShopOwner(input) → signUp + RPC
lib/shop/shopStaffUser.ts	fetchAppStaffUser, resolveShopSession — extended for admin + pending_owner
context/ShopAuthContext.tsx	isAdmin, isPendingOwner, registerOwner, pending_approval login result
components/AppBootstrap.tsx	Admin → /admin; pending → /welcome?pending=1
components/admin/AdminPanel.tsx	Full Super Admin UI
app/admin/_layout.tsx	Stack layout for admin route
supabase/apply-pitstop-2.0-step5-admin-approval.sql	Enum, RLS, RPCs, is_platform_admin()

Context flags quick reference

Flag	When true	Route
isAdmin	users.role = admin	/admin
isPendingOwner	users.role = pending_owner	/welcome (blocked)
isOwner	users.role = owner, active	/shop
isPremium	shops.is_premium = true for owner's shop	/shop — unlocks wash CMS sections
isBranchManager	users.role = branch_manager	/shop (branch filtered)

28. Car Wash Customer UX — Busy · Loyalty · Progress

مميزات تجربة عميل الغسيل — شارة المزدحم، الولاء، وتقديم الحجز

This section documents the three customer-facing Car Wash features added in the latest release, plus the auth fixes that unblock login on web. All three UX features work **offline-first** (AsyncStorage) and do not require new Supabase tables.

28.1 Dynamic Busy Status Indicator

Purpose: When a wash owner is overwhelmed, they set branch status to **Busy**. Customers see a clear red pulsing notice before booking — on the shop list and shop profile.

الغرض: يضبط صاحب المحل حالة «مزدحم» في لوحة التحكم، فيظهر للعميل تنبيه أحمر نابض في قائمة المحلات وصفحة المحل.

Owner side

1. Login as wash owner → /shop → WashOwnerPanel
2. Section **Branch status** (wash_status_* keys): choose Open / Closed / **Busy** / Vacation
3. Save → saveWashBranchStatus() writes to branch storage and calls syncWashBranchToShopExtras()

Data path

```
WashOwnerPanel.saveWashBranchStatus()
  → washBranchStorage (active branch)
  → syncWashBranchToShopExtras()
  → shopExtrasStorage (key per shopId)
  → field: washShopStatus = 'open' | 'closed' | 'busy' | 'vacation'
  → ShopCatalogContext loads extras for customer lists
```

Important fix: normalizeExtras() in shopExtrasStorage.ts now preserves washShopStatus, vacation dates, and activeBranchId — previously these fields could be stripped on save.

Customer UI

Screen	Component	Condition
Shop list card	WashBusyBadge compact	shop.type === 'wash' AND extras.washShopStatus === 'busy'
Shop profile	WashBusyBadge full width	Same condition

Visual: red border/background tint, animated opacity pulse on dot, exclamation icon, bilingual text.

i18n keys

Key	EN (summary)	AR (summary)
wash_busy_customer_notice	Shop is busy — expect longer wait	المحل مزدحم حالياً — قد يكون الانتظار أطول

How to test

1. Owner: set branch status to Busy → Save
2. Customer: open wash shop list in same area → red badge on that shop card
3. Open shop profile → full busy notice appears
4. Owner: set back to Open → badge disappears after catalog refresh

28.2 Offline-First Loyalty Stamps

Purpose: Reward repeat wash customers. Each completed wash booking adds one stamp; after **5 stamps** the customer receives a **100% off coupon** valid 90 days.

الغرض: برنامج ولاء — ختم واحد لكل غسيل مكتمل، وبعد 5 أختام كوبون خصم 100% لمدة 90 يوم.

Rules

- Only shopType === 'wash' bookings count
- Stamp added when booking status becomes done (not on create/cancel)
- Idempotent: each booking.id processed once (processedBookingIds array, max 200 kept)
- At 5 stamps: generate coupon, reset stamps to 0, show reward modal on Home
- Customer key: id:{customerId} or fallback phone:{phone}
- Guest users: no loyalty card (login required)

Storage

AsyncStorage key: @pitstop/loyalty_stamps

```
{
  "id:uuid-or-phone:012...": {
    "stamps": 3,
    "processedBookingIds": ["booking-1", "booking-2"],
    "pendingReward": null,
    "rewards": [{ "code": "LOYAL-ABC123", "discountType": "percent", "discountValue": 100, ... }]
  }
}
```

Code flow

Owner: updateBookingStatus(id, 'done')

- storage.ts calls recordWashBookingDone(booking, previousStatus)
- stamp +1 (or unlock reward at 5)

Customer: Home tab focus

- listBookingsForPhone()
- syncLoyaltyFromBookings() – backfill missed done washes
- getLoyaltyStamps() → LoyaltyCard UI
- consumePendingLoyaltyReward() → modal if new reward

Reward coupon format

Field	Value
code	LOYAL- + 6 random uppercase chars
discountType	percent
discountValue	100 (free wash)
expiresAt	90 days from issue

Limitation: coupon is stored locally; booking checkout does not yet validate LOYAL-* codes automatically — future enhancement.

UI — LoyaltyCard on Home

- 5 car-icon stamp circles; filled stamps use theme accent
- Progress text: "{count} / {goal} stamps"
- Hidden for guests and logged-out users

i18n keys

Key	Usage
loyalty_card_title	Card heading
loyalty_card_lead	Short program description

loyalty_card_progress	Stamp count ({count}, {goal})
loyalty_reward_title	Reward modal title
loyalty_reward_body	Modal body with {code}
loyalty_reward_dismiss	Close button

How to test

1. Login as customer@demo.com
2. Book a wash → owner marks booking Done (5 times, or change dates/status in storage for demo)
3. Home tab: stamp count increases after each done
4. On 5th stamp: modal shows LOYAL-XXXXXX, stamps reset to 0
5. Re-open Home: same booking does not add duplicate stamp

28.3 Live Queue Progress Bar (My Bookings)

Purpose: For *today's* active appointments, replace the static status badge with a 4-step horizontal timeline so customers see where they are in the queue.

الغرض: للحجوزات اليوم فقط — شريط تقدم من 4 خطوات يوضح: قيد الانتظار → مؤكد → جاري التنفيذ → تم.

When shown

```
showTimeline =
  isBookingToday(scheduledAt)  // same calendar day as device
  AND status IN (pending, confirmed, in_progress, done)
```

Cancelled / no-show bookings keep the normal colored status badge only.

Steps (left → right)

Step	Booking status	Visual
1	pending	First circle active
2	confirmed	Connector + circle 2 active
3	in_progress	Step 3 highlighted (bold label)
4	done	All steps accent-colored with check icons

Component: BookingProgressTimeline.tsx — uses theme accent for reached steps, muted for future steps.

i18n keys

Key	Step label
booking_progress_title	Section title above timeline
booking_progress_pending	Pending
booking_progress_confirmed	Confirmed
booking_progress_in_progress	In progress
booking_progress_done	Done

How to test

1. Customer books wash for **today**
2. My Bookings: timeline visible, step 1 active (pending)
3. Owner confirms → customer refreshes tab → step 2 active

4. Owner sets In progress → step 3; Done → all steps complete
5. Book for tomorrow → only status badge, no timeline

28.4 Auth fixes (prerequisite for all roles)

These infrastructure + code fixes were required before customer/owner/admin login worked reliably on web:

Issue	Fix
Email provider disabled in Supabase	Dashboard → Auth → Providers → enable Email
Demo accounts unconfirmed	Auto-confirm on create OR SQL update email_confirmed_at
Web login stuck "Signing in..."	authMutationLock.ts + defer onAuthStateChange work
Browser tabs syncing auth	webTabAuthIsolation.ts — see §31
Admin not routing to panel	ok_admin result → /admin from welcome

Full checklist: [supabase/SETUP-AUTH-CHECKLIST.md](#)

إصلاحات المصادقة: تفعيل Email في Supabase، تأكيد الحسابات التجريبية، ومنع تعليق signIn على الويب.

28.5 File reference (Car Wash UX)

File	Role
components/ui/WashBusyBadge.tsx	Customer busy notice UI
components/ui/ShopListCard.tsx	Shows compact busy badge
app/shop-profile/[shopId].tsx	Shows full busy badge
lib/booking/shopExtrasStorage.ts	Persists washShopStatus
components/customer/LoyaltyCard.tsx	Home stamp card
lib/booking/loyaltyStampsStorage.ts	Stamp logic + coupons
lib/booking/storage.ts	Calls recordWashBookingDone on status update
app/(tabs)/index.tsx	Loyalty card + reward modal
components/customer/BookingProgressTimeline.tsx	4-step progress UI
app/(tabs)/bookings.tsx	Today timeline integration
lib/auth/authMutationLock.ts	Web auth deadlock guard
lib/i18n/strings.ts	All EN/AR strings above

29. SaaS Premium — Free vs Premium (Wash Owner)

اشتراك Premium للمغسلة — مجاني مقابل مدفوع

Wash owners use a **Freemium SaaS** model. Flag: `public.shops.is_premium` → `ShopAuthContext.isPremium`. SQL: `apply-pitstop-2.0-step6-shop-premium.sql`.

29.1 Enable Premium (demo)

```
update public.shops set is_premium = true where id = 'shop-wash-nile';
```

29.2 Free vs Premium

Free ✓	Premium only (locked with gold badge)
Today + pending stats, branch status, profile, gallery, GPS, bookings	Multi-branch tabs, analytics, services, hours, coupons, managers, employees, reviews, PDF reports, extra dashboard stats

29.3 Flow

```
Owner /shop → isPremium?  
  No → PremiumFeatureGate dims sections → tap → PremiumUpgradeModal  
  Yes → full WashOwnerPanel CMS + branch switcher
```

30. Branch GPS & Customer Location Flow

موقع الفرع بالـ Owner — GPS يلتقط Customer يشوف المسافة والاتجاهات

Owner: Profile → Set Branch Location via GPS

- `saveBranchCoordinates()` → `shop_branches` + shops updated in Supabase
- Customer: `/nearby/wash` reloads (`useFocusEffect`) → Haversine sort
- Customer: shop-profile → Directions → Google Maps `?q=lat,lng`

30.1 El Rehab demo

1. Auth user `rehab.wash@demo.com / demo123`
2. Run `seed-el-rehab-wash.sql` — coords `30.0244, 31.4939`
3. Owner captures GPS → customer sees updated distance in Nearby

Modules: `washBranchNearby.ts`, `branchRepository.ts`, `nearby.ts`, `contact.ts`, `washBranchStorage.saveBranchCoordinates`.

31. Web Multi-Tab Auth Isolation

كل تاب متصفح له session مستقل — owner في تاب و customer في تاب

- sessionStorage for Supabase JWT + app session keys
- Disable Supabase BroadcastChannel before createClient
- signOut({ scope: 'local' }) on web logout

Test: Tab1 owner → duplicate → Tab2 customer login → Tab1 stays owner. Hard refresh after code deploy.

32. Wash Pricing — Services Only (No Base Price)

أسعار الغسيل من قائمة الخدمات فقط — بدون سعر أساسي

Removed owner "Base price" from WashOwnerPanel. Customers see per-service prices on shop profile. Booking uses `selectedService.priceEgp` or first active wash service. Maintenance shops still use default service price in generic owner panel.
